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1      *=====*
2      * PROGRAM : BANKING-SYSTEM *
3      * AUTHOR  : BANK OF COBOL - DATA PROCESSING DEPT *
4      * PURPOSE : IN-MEMORY BANKING DATABASE - 1000 CLIENT ACCOUNTS *
5      * OPS     : DEPOSITS AND WITHDRAWALS ONLY *
6      * RULES   : ALL BUSINESS RULES ARE HARDCODED IN WS-RULES *
7      *=====*
8      IDENTIFICATION DIVISION.
9      PROGRAM-ID. BANKING-SYSTEM.
10     AUTHOR. BANK-OF-COBOL-DP.
11     DATE-WRITTEN. 2024-01-15.
12
13     ENVIRONMENT DIVISION.
14     CONFIGURATION SECTION.
15     SOURCE-COMPUTER. IBM-MAINFRAME.
16     OBJECT-COMPUTER. IBM-MAINFRAME.
17
18     DATA DIVISION.
19     WORKING-STORAGE SECTION.
20
21     *=====*
22     * HARDCODED BUSINESS RULES *
23     *=====*
24     01 WS-RULES.
25         05 BR-MIN-BALANCE PIC 9(8)V99 VALUE 100.00.
26         05 BR-MAX-WD-SINGLE PIC 9(8)V99 VALUE 10000.00.
27         05 BR-MAX-DEP-SINGLE PIC 9(8)V99 VALUE 50000.00.
28         05 BR-MAX-WD-DAILY PIC 9(8)V99 VALUE 20000.00.
29         05 BR-MAX-DEP-DAILY PIC 9(8)V99 VALUE 100000.00.
30         05 BR-MIN-TX-AMOUNT PIC 9(8)V99 VALUE 1.00.
31         05 BR-INITIAL-BALANCE PIC 9(8)V99 VALUE 500.00.
32         05 BR-VIP-THRESHOLD PIC 9(8)V99 VALUE 50000.00.
33         05 BR-VIP-MAX-WD-SINGLE PIC 9(8)V99 VALUE 50000.00.
34         05 BR-VIP-MAX-DEP-SINGLE PIC 9(8)V99 VALUE 250000.00.
35         05 BR-VIP-MAX-WD-DAILY PIC 9(8)V99 VALUE 100000.00.
36         05 BR-VIP-MAX-DEP-DAILY PIC 9(8)V99 VALUE 500000.00.
37         05 BR-MAX-CLIENTS PIC 9(4) VALUE 1000.
38         05 BR-MAX-TX-LOG PIC 9(5) VALUE 5000.
39         05 BR-FIRST-CLIENT-ID PIC 9(6) VALUE 100001.
40
41     *=====*
42     * CLIENT DATABASE - 1000 ACCOUNT RECORDS (IN-MEMORY) *
43     *=====*
44     01 WS-CLIENT-DB.
45         05 DB-CLIENT-COUNT PIC 9(4) VALUE 0.
46         05 DB-ENTRY OCCURS 1000 TIMES
47             INDEXED BY DB-IDX.
48             10 DB-CLIENT-ID PIC 9(6).
49             10 DB-LAST-NAME PIC X(15).
50             10 DB-FIRST-NAME PIC X(10).
51             10 DB-BALANCE PIC S9(10)V99.
52             10 DB-STATUS PIC X(1).
53             88 DB-ACTIVE VALUE 'A'.
54             88 DB-FROZEN VALUE 'F'.
55             88 DB-CLOSED VALUE 'C'.
56             10 DB-ACCT-TYPE PIC X(1).
57             88 DB-REGULAR VALUE 'R'.
58             88 DB-VIP VALUE 'V'.
59             10 DB-DAILY-WD PIC 9(10)V99.
60             10 DB-DAILY-DEP PIC 9(10)V99.
61             10 DB-OPEN-DATE PIC X(10).
62             10 DB-TX-COUNT PIC 9(6).
63             10 DB-LAST-TX-DATE PIC X(10).
64             10 DB-LAST-TX-TYPE PIC X(1).
65             10 DB-LAST-TX-AMT PIC 9(10)V99.
66
67     *=====*
68     * TRANSACTION LOG - CIRCULAR BUFFER (5000 ENTRIES) *
69     *=====*
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70      01 WS-TX-LOG.
71          05 TX-TOTAL-COUNT          PIC 9(5)  VALUE 0.
72          05 TX-NEXT-ID              PIC 9(8)  VALUE 1.
73          05 TX-ENTRY OCCURS 5000 TIMES
74              INDEXED BY TX-IDX.
75              10 TX-ID                PIC 9(8).
76              10 TX-DATE              PIC X(10).
77              10 TX-TIME              PIC X(8).
78              10 TX-CLIENT-ID         PIC 9(6).
79              10 TX-TYPE              PIC X(1).
80              88 TX-DEPOSIT           VALUE 'D'.
81              88 TX-WITHDRAWAL        VALUE 'W'.
82              10 TX-AMOUNT            PIC 9(10)V99.
83              10 TX-BAL-AFTER         PIC S9(10)V99.
84              10 TX-RESULT            PIC X(2).
85              88 TX-APPROVED          VALUE 'OK'.
86              88 TX-REJECTED          VALUE 'RJ'.
87              10 TX-REJECT-REASON     PIC X(30).
88
89      *=====*
90      *  WORK FIELDS                                *
91      *=====*
92      01 WS-WORK.
93          05 WW-SEARCH-ID             PIC 9(6)    VALUE 0.
94          05 WW-FOUND-IDX             PIC 9(4)    VALUE 0.
95          05 WW-FOUND-FLAG            PIC X(1)    VALUE 'N'.
96          05 WW-AMOUNT                PIC 9(10)V99 VALUE 0.
97          05 WW-NEW-BALANCE           PIC S9(10)V99 VALUE 0.
98          05 WW-RESULT                PIC X(2)    VALUE SPACES.
99          05 WW-ERROR-MSG             PIC X(60)   VALUE SPACES.
100         05 WW-MENU-CHOICE            PIC X(1)    VALUE SPACES.
101         05 WW-RUN-FLAG               PIC X(1)    VALUE 'Y'.
102         05 WW-LOOP-CTR               PIC 9(4)    VALUE 0.
103         05 WW-INIT-CTR               PIC 9(4)    VALUE 0.
104         05 WW-LN-SEED                PIC 9(2)    VALUE 0.
105         05 WW-FN-SEED                PIC 9(2)    VALUE 0.
106         05 WW-BAL-SEED               PIC 9(10)V99 VALUE 0.
107         05 WW-MOD-RESULT              PIC 9(4)    VALUE 0.
108         05 WW-TX-SLOT                PIC 9(5)    VALUE 0.
109         05 WW-AVAIL-WD               PIC 9(10)V99 VALUE 0.
110         05 WW-AVAIL-DEP              PIC 9(10)V99 VALUE 0.
111         05 WW-CURRENT-OP             PIC X(1)    VALUE SPACES.
112         05 WW-DATE-SIM               PIC X(10)   VALUE '2024-01-15'.
113         05 WW-TIME-SIM               PIC X(8)    VALUE '09:00:00'.
114         05 WW-TOTAL-DEP-DAY          PIC 9(12)V99 VALUE 0.
115         05 WW-TOTAL-WD-DAY           PIC 9(12)V99 VALUE 0.
116         05 WW-PAGE-LIMIT             PIC 9(4)    VALUE 20.
117         05 WW-PAGE-START             PIC 9(4)    VALUE 1.
118         05 WW-PAGE-END               PIC 9(4)    VALUE 0.
119         05 WW-STATUS-LABEL           PIC X(8)    VALUE SPACES.
120         05 WW-TYPE-LABEL             PIC X(8)    VALUE SPACES.
121
122      01 WS-DISPLAY.
123          05 WD-BALANCE                PIC $$$,$$$,$$9.99.
124          05 WD-AMOUNT                 PIC $$$,$$$,$$9.99.
125          05 WD-DAILY-WD               PIC $$$,$$$,$$9.99.
126          05 WD-DAILY-DEP             PIC $$$,$$$,$$9.99.
127          05 WD-SEP-LINE               PIC X(65)  VALUE ALL '='.
128          05 WD-DASH-LINE             PIC X(65)  VALUE ALL '- '.
129
130      *=====*
131      *  NAME SEED TABLES FOR PROCEDURAL CLIENT GENERATION  *
132      *=====*
133      01 WS-LNAME-TABLE.
134          05 LT-ENTRY OCCURS 20 TIMES.
135              10 LT-VALUE              PIC X(15).
136
137      01 WS-FNAME-TABLE.
138          05 FT-ENTRY OCCURS 20 TIMES.

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139          10  FT-VALUE          PIC X(10).
140
141      *=====*
142      PROCEDURE DIVISION.
143      *=====*
144
145      *-----*
146      0000-MAIN.
147          PERFORM 1000-INITIALIZE
148          PERFORM 2000-MAIN-MENU
149          UNTIL WW-RUN-FLAG = 'N'
150          PERFORM 9999-END-PROGRAM
151          STOP RUN
152      .
153
154      *=====*
155      *  INITIALIZATION                      *
156      *=====*
157
158      1000-INITIALIZE.
159          PERFORM 1100-LOAD-NAME-TABLES
160          PERFORM 1200-POPULATE-CLIENTS
161          PERFORM 1300-DISPLAY-WELCOME
162      .
163
164      *-----*
165      1100-LOAD-NAME-TABLES.
166          MOVE 'SMITH      ' TO LT-VALUE(1)
167          MOVE 'JOHNSON    ' TO LT-VALUE(2)
168          MOVE 'WILLIAMS   ' TO LT-VALUE(3)
169          MOVE 'JONES      ' TO LT-VALUE(4)
170          MOVE 'BROWN     ' TO LT-VALUE(5)
171          MOVE 'DAVIS      ' TO LT-VALUE(6)
172          MOVE 'MILLER     ' TO LT-VALUE(7)
173          MOVE 'WILSON     ' TO LT-VALUE(8)
174          MOVE 'MOORE      ' TO LT-VALUE(9)
175          MOVE 'TAYLOR     ' TO LT-VALUE(10)
176          MOVE 'ANDERSON   ' TO LT-VALUE(11)
177          MOVE 'THOMAS     ' TO LT-VALUE(12)
178          MOVE 'JACKSON    ' TO LT-VALUE(13)
179          MOVE 'WHITE      ' TO LT-VALUE(14)
180          MOVE 'HARRIS     ' TO LT-VALUE(15)
181          MOVE 'MARTIN     ' TO LT-VALUE(16)
182          MOVE 'GARCIA     ' TO LT-VALUE(17)
183          MOVE 'MARTINEZ   ' TO LT-VALUE(18)
184          MOVE 'ROBINSON   ' TO LT-VALUE(19)
185          MOVE 'CLARK      ' TO LT-VALUE(20)
186          MOVE 'JAMES      ' TO FT-VALUE(1)
187          MOVE 'JOHN       ' TO FT-VALUE(2)
188          MOVE 'ROBERT     ' TO FT-VALUE(3)
189          MOVE 'MICHAEL    ' TO FT-VALUE(4)
190          MOVE 'WILLIAM    ' TO FT-VALUE(5)
191          MOVE 'DAVID      ' TO FT-VALUE(6)
192          MOVE 'RICHARD    ' TO FT-VALUE(7)
193          MOVE 'JOSEPH     ' TO FT-VALUE(8)
194          MOVE 'THOMAS     ' TO FT-VALUE(9)
195          MOVE 'CHARLES    ' TO FT-VALUE(10)
196          MOVE 'MARY       ' TO FT-VALUE(11)
197          MOVE 'PATRICIA   ' TO FT-VALUE(12)
198          MOVE 'LINDA      ' TO FT-VALUE(13)
199          MOVE 'BARBARA    ' TO FT-VALUE(14)
200          MOVE 'ELIZABETH ' TO FT-VALUE(15)
201          MOVE 'JENNIFER   ' TO FT-VALUE(16)
202          MOVE 'MARIA      ' TO FT-VALUE(17)
203          MOVE 'SUSAN      ' TO FT-VALUE(18)
204          MOVE 'MARGARET   ' TO FT-VALUE(19)
205          MOVE 'DOROTHY    ' TO FT-VALUE(20)
206      .
207

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208      *-----*
209      1200-POPULATE-CLIENTS.
210          MOVE BR-MAX-CLIENTS TO DB-CLIENT-COUNT
211          PERFORM VARYING WW-INIT-CTR FROM 1 BY 1
212              UNTIL WW-INIT-CTR > 1000
213              PERFORM 1210-INIT-ONE-CLIENT
214          END-PERFORM
215      .
216
217      *-----*
218      1210-INIT-ONE-CLIENT.
219          COMPUTE DB-CLIENT-ID(WW-INIT-CTR) = 100000 + WW-INIT-CTR
220          COMPUTE WW-LN-SEED =
221              FUNCTION MOD(WW-INIT-CTR - 1, 20) + 1
222          MOVE LT-VALUE(WW-LN-SEED) TO DB-LAST-NAME(WW-INIT-CTR)
223          COMPUTE WW-FN-SEED =
224              FUNCTION MOD(WW-INIT-CTR + 6, 20) + 1
225          MOVE FT-VALUE(WW-FN-SEED) TO DB-FIRST-NAME(WW-INIT-CTR)
226          COMPUTE WW-BAL-SEED = 500.00 + (WW-INIT-CTR * 99.50)
227          MOVE WW-BAL-SEED TO DB-BALANCE(WW-INIT-CTR)
228          MOVE WW-DATE-SIM TO DB-OPEN-DATE(WW-INIT-CTR)
229          MOVE ZEROS TO DB-DAILY-WD(WW-INIT-CTR)
230          MOVE ZEROS TO DB-DAILY-DEP(WW-INIT-CTR)
231          MOVE ZEROS TO DB-TX-COUNT(WW-INIT-CTR)
232          MOVE WW-DATE-SIM TO DB-LAST-TX-DATE(WW-INIT-CTR)
233          MOVE SPACES TO DB-LAST-TX-TYPE(WW-INIT-CTR)
234          MOVE ZEROS TO DB-LAST-TX-AMT(WW-INIT-CTR)
235          COMPUTE WW-MOD-RESULT = FUNCTION MOD(WW-INIT-CTR, 100)
236          EVALUATE TRUE
237              WHEN WW-MOD-RESULT = 0
238                  MOVE 'F' TO DB-STATUS(WW-INIT-CTR)
239              WHEN OTHER
240                  COMPUTE WW-MOD-RESULT =
241                      FUNCTION MOD(WW-INIT-CTR, 50)
242                  IF WW-MOD-RESULT = 0
243                      MOVE 'C' TO DB-STATUS(WW-INIT-CTR)
244                  ELSE
245                      MOVE 'A' TO DB-STATUS(WW-INIT-CTR)
246              END-IF
247          END-EVALUATE
248          IF DB-BALANCE(WW-INIT-CTR) > BR-VIP-THRESHOLD
249              MOVE 'V' TO DB-ACCT-TYPE(WW-INIT-CTR)
250          ELSE
251              MOVE 'R' TO DB-ACCT-TYPE(WW-INIT-CTR)
252          END-IF
253      .
254
255      *-----*
256      1300-DISPLAY-WELCOME.
257          DISPLAY WD-SEP-LINE
258          DISPLAY '  BANK OF COBOL  -  CORE BANKING SYSTEM  v1.0 '
259          DISPLAY '  ESTABLISHED 1959  -  MAINFRAME DIVISION      '
260          DISPLAY WD-SEP-LINE
261          DISPLAY '  DATABASE LOADED:  ' DB-CLIENT-COUNT
262              '  CLIENT ACCOUNTS'
263          DISPLAY '  AUTHORIZED OPS :  DEPOSIT  |  WITHDRAWAL '
264          DISPLAY WD-SEP-LINE
265      .
266
267      *=====*
268      *  MAIN MENU                                     *
269      *=====*
270
271      2000-MAIN-MENU.
272          DISPLAY ' '
273          DISPLAY WD-DASH-LINE
274          DISPLAY '  MAIN MENU'
275          DISPLAY WD-DASH-LINE
276          DISPLAY '  1. DEPOSIT'

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277      DISPLAY ' 2. WITHDRAWAL '
278      DISPLAY ' 3. BALANCE INQUIRY '
279      DISPLAY ' 4. CLIENT ACCOUNT DETAIL '
280      DISPLAY ' 5. LIST CLIENTS (20 AT A TIME) '
281      DISPLAY ' 6. DAILY SUMMARY REPORT '
282      DISPLAY ' 7. RESET DAILY LIMITS (END OF DAY) '
283      DISPLAY ' 8. TRANSACTION LOG (LAST 20) '
284      DISPLAY ' X. EXIT '
285      DISPLAY WD-DASH-LINE
286      DISPLAY ' CHOICE: ' WITH NO ADVANCING
287      ACCEPT WW-MENU-CHOICE
288      EVALUATE WW-MENU-CHOICE
289          WHEN '1'
290              PERFORM 3000-PROCESS-DEPOSIT
291          WHEN '2'
292              PERFORM 4000-PROCESS-WITHDRAWAL
293          WHEN '3'
294              PERFORM 5000-BALANCE-INQUIRY
295          WHEN '4'
296              PERFORM 6000-ACCOUNT-DETAIL
297          WHEN '5'
298              PERFORM 7000-LIST-CLIENTS
299          WHEN '6'
300              PERFORM 8000-DAILY-SUMMARY
301          WHEN '7'
302              PERFORM 9500-RESET-DAILY-LIMITS
303          WHEN '8'
304              PERFORM 9600-SHOW-TX-LOG
305          WHEN 'X'
306              MOVE 'N' TO WW-RUN-FLAG
307          WHEN 'X'
308              MOVE 'N' TO WW-RUN-FLAG
309          WHEN OTHER
310              DISPLAY ' *** INVALID CHOICE *** '
311      END-EVALUATE
312      .
313
314      *=====*
315      * DEPOSIT PROCESSING *
316      *=====*
317
318      3000-PROCESS-DEPOSIT.
319      MOVE 'D' TO WW-CURRENT-OP
320      DISPLAY WD-DASH-LINE
321      DISPLAY ' DEPOSIT '
322      DISPLAY WD-DASH-LINE
323      DISPLAY ' CLIENT ID (0=CANCEL): ' WITH NO ADVANCING
324      ACCEPT WW-SEARCH-ID
325      IF WW-SEARCH-ID = 0
326          DISPLAY ' OPERATION CANCELLED. '
327      EXIT PARAGRAPH
328      END-IF
329      PERFORM 9000-FIND-CLIENT
330      IF WW-FOUND-FLAG = 'N'
331          DISPLAY ' *** CLIENT NOT FOUND: ' WW-SEARCH-ID
332      EXIT PARAGRAPH
333      END-IF
334      PERFORM 9100-SHOW-CLIENT-HEADER
335      MOVE DB-BALANCE(WW-FOUND-IDX) TO WD-BALANCE
336      DISPLAY ' CURRENT BALANCE : ' WD-BALANCE
337      DISPLAY ' AMOUNT (0=CANCEL): ' WITH NO ADVANCING
338      ACCEPT WW-AMOUNT
339      IF WW-AMOUNT = 0
340          DISPLAY ' OPERATION CANCELLED. '
341      EXIT PARAGRAPH
342      END-IF
343      PERFORM 3100-VALIDATE-DEPOSIT
344      IF WW-RESULT = 'OK'
345          PERFORM 3200-APPLY-DEPOSIT

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346      ELSE
347          DISPLAY ' *** REJECTED: ' WW-ERROR-MSG
348      END-IF
349      PERFORM 9200-LOG-TRANSACTION
350      .
351
352      *-----*
353      3100-VALIDATE-DEPOSIT.
354          MOVE 'OK' TO WW-RESULT
355          MOVE SPACES TO WW-ERROR-MSG
356          IF DB-FROZEN(WW-FOUND-IDX)
357              MOVE 'RJ' TO WW-RESULT
358              MOVE 'ACCOUNT FROZEN - VISIT BRANCH TO UNFREEZE'
359                  TO WW-ERROR-MSG
360          EXIT PARAGRAPH
361      END-IF
362      IF DB-CLOSED(WW-FOUND-IDX)
363          MOVE 'RJ' TO WW-RESULT
364          MOVE 'ACCOUNT IS PERMANENTLY CLOSED'
365              TO WW-ERROR-MSG
366          EXIT PARAGRAPH
367      END-IF
368      IF WW-AMOUNT < BR-MIN-TX-AMOUNT
369          MOVE 'RJ' TO WW-RESULT
370          MOVE 'AMOUNT BELOW MINIMUM DEPOSIT OF $1.00'
371              TO WW-ERROR-MSG
372          EXIT PARAGRAPH
373      END-IF
374      IF DB-VIP(WW-FOUND-IDX)
375          IF WW-AMOUNT > BR-VIP-MAX-DEP-SINGLE
376              MOVE 'RJ' TO WW-RESULT
377              MOVE 'EXCEEDS VIP SINGLE DEPOSIT CAP $250,000.00'
378                  TO WW-ERROR-MSG
379          EXIT PARAGRAPH
380      END-IF
381      ELSE
382          IF WW-AMOUNT > BR-MAX-DEP-SINGLE
383              MOVE 'RJ' TO WW-RESULT
384              MOVE 'EXCEEDS REGULAR SINGLE DEPOSIT CAP $50,000'
385                  TO WW-ERROR-MSG
386          EXIT PARAGRAPH
387      END-IF
388      END-IF
389      COMPUTE WW-AVAIL-DEP =
390          DB-DAILY-DEP(WW-FOUND-IDX) + WW-AMOUNT
391      IF DB-VIP(WW-FOUND-IDX)
392          IF WW-AVAIL-DEP > BR-VIP-MAX-DEP-DAILY
393              MOVE 'RJ' TO WW-RESULT
394              MOVE 'VIP DAILY DEPOSIT LIMIT REACHED $500,000'
395                  TO WW-ERROR-MSG
396          EXIT PARAGRAPH
397      END-IF
398      ELSE
399          IF WW-AVAIL-DEP > BR-MAX-DEP-DAILY
400              MOVE 'RJ' TO WW-RESULT
401              MOVE 'DAILY DEPOSIT LIMIT REACHED $100,000.00'
402                  TO WW-ERROR-MSG
403          EXIT PARAGRAPH
404      END-IF
405      END-IF
406      .
407
408      *-----*
409      3200-APPLY-DEPOSIT.
410          COMPUTE DB-BALANCE(WW-FOUND-IDX) =
411              DB-BALANCE(WW-FOUND-IDX) + WW-AMOUNT
412          ADD WW-AMOUNT TO DB-DAILY-DEP(WW-FOUND-IDX)
413          ADD 1 TO DB-TX-COUNT(WW-FOUND-IDX)
414          MOVE WW-DATE-SIM TO DB-LAST-TX-DATE(WW-FOUND-IDX)

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415     MOVE 'D' TO DB-LAST-TX-TYPE(WW-FOUND-IDX)
416     MOVE WW-AMOUNT TO DB-LAST-TX-AMT(WW-FOUND-IDX)
417     IF DB-BALANCE(WW-FOUND-IDX) > BR-VIP-THRESHOLD
418         MOVE 'V' TO DB-ACCT-TYPE(WW-FOUND-IDX)
419     END-IF
420     MOVE WW-AMOUNT TO WD-AMOUNT
421     MOVE DB-BALANCE(WW-FOUND-IDX) TO WD-BALANCE
422     DISPLAY ' >> DEPOSIT APPROVED <<'
423     DISPLAY '   AMOUNT DEPOSITED : ' WD-AMOUNT
424     DISPLAY '   NEW BALANCE       : ' WD-BALANCE
425     .
426
427     *=====*
428     *   WITHDRAWAL PROCESSING                                   *
429     *=====*
430
431     4000-PROCESS-WITHDRAWAL.
432     MOVE 'W' TO WW-CURRENT-OP
433     DISPLAY WD-DASH-LINE
434     DISPLAY '   WITHDRAWAL '
435     DISPLAY WD-DASH-LINE
436     DISPLAY '   CLIENT ID (0=CANCEL): ' WITH NO ADVANCING
437     ACCEPT WW-SEARCH-ID
438     IF WW-SEARCH-ID = 0
439         DISPLAY '   OPERATION CANCELLED.'
440     EXIT PARAGRAPH
441     END-IF
442     PERFORM 9000-FIND-CLIENT
443     IF WW-FOUND-FLAG = 'N'
444         DISPLAY '   *** CLIENT NOT FOUND: ' WW-SEARCH-ID
445     EXIT PARAGRAPH
446     END-IF
447     PERFORM 9100-SHOW-CLIENT-HEADER
448     MOVE DB-BALANCE(WW-FOUND-IDX) TO WD-BALANCE
449     DISPLAY '   CURRENT BALANCE : ' WD-BALANCE
450     DISPLAY '   AMOUNT (0=CANCEL): ' WITH NO ADVANCING
451     ACCEPT WW-AMOUNT
452     IF WW-AMOUNT = 0
453         DISPLAY '   OPERATION CANCELLED.'
454     EXIT PARAGRAPH
455     END-IF
456     PERFORM 4100-VALIDATE-WITHDRAWAL
457     IF WW-RESULT = 'OK'
458         PERFORM 4200-APPLY-WITHDRAWAL
459     ELSE
460         DISPLAY '   *** REJECTED: ' WW-ERROR-MSG
461     END-IF
462     PERFORM 9200-LOG-TRANSACTION
463     .
464
465     *-----*
466     4100-VALIDATE-WITHDRAWAL.
467     MOVE 'OK' TO WW-RESULT
468     MOVE SPACES TO WW-ERROR-MSG
469     IF DB-FROZEN(WW-FOUND-IDX)
470         MOVE 'RJ' TO WW-RESULT
471         MOVE 'ACCOUNT FROZEN - VISIT BRANCH TO UNFREEZE'
472             TO WW-ERROR-MSG
473     EXIT PARAGRAPH
474     END-IF
475     IF DB-CLOSED(WW-FOUND-IDX)
476         MOVE 'RJ' TO WW-RESULT
477         MOVE 'ACCOUNT IS PERMANENTLY CLOSED'
478             TO WW-ERROR-MSG
479     EXIT PARAGRAPH
480     END-IF
481     IF WW-AMOUNT < BR-MIN-TX-AMOUNT
482         MOVE 'RJ' TO WW-RESULT
483         MOVE 'AMOUNT BELOW MINIMUM WITHDRAWAL OF $1.00'

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484             TO WW-ERROR-MSG
485         EXIT PARAGRAPH
486     END-IF
487     IF DB-VIP(WW-FOUND-IDX)
488         IF WW-AMOUNT > BR-VIP-MAX-WD-SINGLE
489             MOVE 'RJ' TO WW-RESULT
490             MOVE 'EXCEEDS VIP SINGLE WITHDRAWAL CAP $50,000'
491                 TO WW-ERROR-MSG
492             EXIT PARAGRAPH
493         END-IF
494     ELSE
495         IF WW-AMOUNT > BR-MAX-WD-SINGLE
496             MOVE 'RJ' TO WW-RESULT
497             MOVE 'EXCEEDS REGULAR SINGLE WITHDRAWAL CAP $10K'
498                 TO WW-ERROR-MSG
499             EXIT PARAGRAPH
500         END-IF
501     END-IF
502     COMPUTE WW-AVAIL-WD =
503         DB-DAILY-WD(WW-FOUND-IDX) + WW-AMOUNT
504     IF DB-VIP(WW-FOUND-IDX)
505         IF WW-AVAIL-WD > BR-VIP-MAX-WD-DAILY
506             MOVE 'RJ' TO WW-RESULT
507             MOVE 'VIP DAILY WITHDRAWAL LIMIT REACHED $100,000'
508                 TO WW-ERROR-MSG
509             EXIT PARAGRAPH
510         END-IF
511     ELSE
512         IF WW-AVAIL-WD > BR-MAX-WD-DAILY
513             MOVE 'RJ' TO WW-RESULT
514             MOVE 'DAILY WITHDRAWAL LIMIT REACHED $20,000.00'
515                 TO WW-ERROR-MSG
516             EXIT PARAGRAPH
517         END-IF
518     END-IF
519     COMPUTE WW-NEW-BALANCE =
520         DB-BALANCE(WW-FOUND-IDX) - WW-AMOUNT
521     IF WW-NEW-BALANCE < BR-MIN-BALANCE
522         MOVE 'RJ' TO WW-RESULT
523         MOVE 'WOULD BREACH MINIMUM REQUIRED BALANCE $100.00'
524             TO WW-ERROR-MSG
525         EXIT PARAGRAPH
526     END-IF
527     IF WW-NEW-BALANCE < 0
528         MOVE 'RJ' TO WW-RESULT
529         MOVE 'INSUFFICIENT FUNDS - OVERDRAFT NOT PERMITTED'
530             TO WW-ERROR-MSG
531         EXIT PARAGRAPH
532     END-IF
533     .
534
535     *-----*
536     4200-APPLY-WITHDRAWAL.
537         COMPUTE DB-BALANCE(WW-FOUND-IDX) =
538             DB-BALANCE(WW-FOUND-IDX) - WW-AMOUNT
539         ADD WW-AMOUNT TO DB-DAILY-WD(WW-FOUND-IDX)
540         ADD 1 TO DB-TX-COUNT(WW-FOUND-IDX)
541         MOVE WW-DATE-SIM TO DB-LAST-TX-DATE(WW-FOUND-IDX)
542         MOVE 'W' TO DB-LAST-TX-TYPE(WW-FOUND-IDX)
543         MOVE WW-AMOUNT TO DB-LAST-TX-AMT(WW-FOUND-IDX)
544         IF DB-VIP(WW-FOUND-IDX)
545             IF DB-BALANCE(WW-FOUND-IDX) < BR-VIP-THRESHOLD
546                 MOVE 'R' TO DB-ACCT-TYPE(WW-FOUND-IDX)
547                 DISPLAY ' NOTE: ACCOUNT DOWNGRADED VIP -> REGULAR'
548             END-IF
549         END-IF
550         MOVE WW-AMOUNT TO WD-AMOUNT
551         MOVE DB-BALANCE(WW-FOUND-IDX) TO WD-BALANCE
552         DISPLAY ' >> WITHDRAWAL APPROVED <<'

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553      DISPLAY '  AMOUNT WITHDRAWN : ' WD-AMOUNT
554      DISPLAY '  NEW BALANCE       : ' WD-BALANCE
555      .
556
557      *=====*
558      *  BALANCE INQUIRY                                *
559      *=====*
560
561      5000-BALANCE-INQUIRY.
562      DISPLAY WD-DASH-LINE
563      DISPLAY '  BALANCE INQUIRY'
564      DISPLAY WD-DASH-LINE
565      DISPLAY '  CLIENT ID (0=CANCEL): ' WITH NO ADVANCING
566      ACCEPT WW-SEARCH-ID
567      IF WW-SEARCH-ID = 0
568          DISPLAY '  CANCELLED.'
569          EXIT PARAGRAPH
570      END-IF
571      PERFORM 9000-FIND-CLIENT
572      IF WW-FOUND-FLAG = 'N'
573          DISPLAY '  *** CLIENT NOT FOUND: ' WW-SEARCH-ID
574          EXIT PARAGRAPH
575      END-IF
576      PERFORM 9100-SHOW-CLIENT-HEADER
577      MOVE DB-BALANCE(WW-FOUND-IDX) TO WD-BALANCE
578      DISPLAY '  CURRENT BALANCE : ' WD-BALANCE
579      MOVE DB-DAILY-WD(WW-FOUND-IDX) TO WD-DAILY-WD
580      DISPLAY '  WITHDRAWN TODAY : ' WD-DAILY-WD
581      MOVE DB-DAILY-DEP(WW-FOUND-IDX) TO WD-DAILY-DEP
582      DISPLAY '  DEPOSITED TODAY : ' WD-DAILY-DEP
583      DISPLAY '  TOTAL TX ON ACCT : '
584      DB-TX-COUNT(WW-FOUND-IDX)
585      .
586
587      *=====*
588      *  ACCOUNT DETAIL                                    *
589      *=====*
590
591      6000-ACCOUNT-DETAIL.
592      DISPLAY WD-DASH-LINE
593      DISPLAY '  ACCOUNT DETAIL'
594      DISPLAY WD-DASH-LINE
595      DISPLAY '  CLIENT ID (0=CANCEL): ' WITH NO ADVANCING
596      ACCEPT WW-SEARCH-ID
597      IF WW-SEARCH-ID = 0
598          DISPLAY '  CANCELLED.'
599          EXIT PARAGRAPH
600      END-IF
601      PERFORM 9000-FIND-CLIENT
602      IF WW-FOUND-FLAG = 'N'
603          DISPLAY '  *** CLIENT NOT FOUND: ' WW-SEARCH-ID
604          EXIT PARAGRAPH
605      END-IF
606      PERFORM 9300-DISPLAY-FULL-RECORD
607      .
608
609      *=====*
610      *  CLIENT LIST                                        *
611      *=====*
612
613      7000-LIST-CLIENTS.
614      DISPLAY WD-DASH-LINE
615      DISPLAY '  CLIENT LIST (PAGE: 20 RECORDS)'
616      DISPLAY WD-DASH-LINE
617      DISPLAY '  START FROM RECORD # (1-1000): '
618      WITH NO ADVANCING
619      ACCEPT WW-PAGE-START
620      IF WW-PAGE-START < 1 OR WW-PAGE-START > 1000
621          MOVE 1 TO WW-PAGE-START

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622      END-IF
623      COMPUTE WW-PAGE-END = WW-PAGE-START + WW-PAGE-LIMIT - 1
624      IF WW-PAGE-END > DB-CLIENT-COUNT
625          MOVE DB-CLIENT-COUNT TO WW-PAGE-END
626      END-IF
627      DISPLAY ' '
628      DISPLAY ' ' IDX CLIENT-ID LAST NAME FIRST'
629      ' BALANCE ST TY'
630      DISPLAY WD-DASH-LINE
631      PERFORM VARYING WW-LOOP-CTR
632          FROM WW-PAGE-START BY 1
633          UNTIL WW-LOOP-CTR > WW-PAGE-END
634          MOVE DB-BALANCE(WW-LOOP-CTR) TO WD-BALANCE
635          DISPLAY ' '
636          WW-LOOP-CTR ' '
637          DB-CLIENT-ID(WW-LOOP-CTR) ' '
638          DB-LAST-NAME(WW-LOOP-CTR)
639          DB-FIRST-NAME(WW-LOOP-CTR) ' '
640          WD-BALANCE ' '
641          DB-STATUS(WW-LOOP-CTR) ' '
642          DB-ACCT-TYPE(WW-LOOP-CTR)
643      END-PERFORM
644      DISPLAY WD-DASH-LINE
645      DISPLAY ' RECORDS ' WW-PAGE-START ' TO '
646          WW-PAGE-END ' OF ' DB-CLIENT-COUNT
647      .
648
649      *=====*
650      * DAILY SUMMARY REPORT *
651      *=====*
652
653      8000-DAILY-SUMMARY.
654      DISPLAY WD-DASH-LINE
655      DISPLAY ' DAILY SUMMARY - ' WW-DATE-SIM
656      DISPLAY WD-DASH-LINE
657      PERFORM 8100-CALC-TOTALS
658      DISPLAY WD-DASH-LINE
659      PERFORM 8200-SHOW-STATUS-COUNTS
660      .
661
662      *-----*
663      8100-CALC-TOTALS.
664      MOVE ZEROS TO WW-TOTAL-DEP-DAY
665      MOVE ZEROS TO WW-TOTAL-WD-DAY
666      PERFORM VARYING WW-LOOP-CTR FROM 1 BY 1
667          UNTIL WW-LOOP-CTR > DB-CLIENT-COUNT
668          ADD DB-DAILY-DEP(WW-LOOP-CTR) TO WW-TOTAL-DEP-DAY
669          ADD DB-DAILY-WD(WW-LOOP-CTR) TO WW-TOTAL-WD-DAY
670      END-PERFORM
671      MOVE WW-TOTAL-DEP-DAY TO WD-DAILY-DEP
672      MOVE WW-TOTAL-WD-DAY TO WD-DAILY-WD
673      DISPLAY ' TOTAL DEPOSITS TODAY : ' WD-DAILY-DEP
674      DISPLAY ' TOTAL WITHDRAWALS TODAY : ' WD-DAILY-WD
675      DISPLAY ' TOTAL TRANSACTIONS LOGGED: ' TX-TOTAL-COUNT
676      .
677
678      *-----*
679      8200-SHOW-STATUS-COUNTS.
680      MOVE ZEROS TO WW-LOOP-CTR
681      MOVE 0 TO WW-AVAIL-DEP
682      MOVE 0 TO WW-AVAIL-WD
683      MOVE 0 TO WW-NEW-BALANCE
684      PERFORM VARYING WW-LOOP-CTR FROM 1 BY 1
685          UNTIL WW-LOOP-CTR > DB-CLIENT-COUNT
686          EVALUATE DB-STATUS(WW-LOOP-CTR)
687              WHEN 'A'
688                  ADD 1 TO WW-AVAIL-DEP
689              WHEN 'F'
690                  ADD 1 TO WW-AVAIL-WD

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691             WHEN 'C'
692                 ADD 1 TO WW-NEW-BALANCE
693             END-EVALUATE
694         END-PERFORM
695         DISPLAY '   ACTIVE ACCOUNTS       : ' WW-AVAIL-DEP
696         DISPLAY '   FROZEN ACCOUNTS       : ' WW-AVAIL-WD
697         DISPLAY '   CLOSED ACCOUNTS      : ' WW-NEW-BALANCE
698         .
699
700         *=====*
701         *  SHARED SUBROUTINES                                     *
702         *=====*
703
704         9000-FIND-CLIENT.
705             MOVE 'N' TO WW-FOUND-FLAG
706             MOVE 0 TO WW-FOUND-IDX
707             PERFORM VARYING WW-LOOP-CTR FROM 1 BY 1
708                 UNTIL WW-LOOP-CTR > DB-CLIENT-COUNT
709                     OR WW-FOUND-FLAG = 'Y'
710                 IF DB-CLIENT-ID(WW-LOOP-CTR) = WW-SEARCH-ID
711                     MOVE 'Y' TO WW-FOUND-FLAG
712                     MOVE WW-LOOP-CTR TO WW-FOUND-IDX
713             END-IF
714         END-PERFORM
715         .
716
717         *-----*
718         9100-SHOW-CLIENT-HEADER.
719             EVALUATE DB-STATUS(WW-FOUND-IDX)
720                 WHEN 'A' MOVE 'ACTIVE ' TO WW-STATUS-LABEL
721                 WHEN 'F' MOVE 'FROZEN ' TO WW-STATUS-LABEL
722                 WHEN 'C' MOVE 'CLOSED ' TO WW-STATUS-LABEL
723                 WHEN OTHER MOVE '?' ' TO WW-STATUS-LABEL
724             END-EVALUATE
725             EVALUATE DB-ACCT-TYPE(WW-FOUND-IDX)
726                 WHEN 'V' MOVE 'VIP ' TO WW-TYPE-LABEL
727                 WHEN 'R' MOVE 'REGULAR ' TO WW-TYPE-LABEL
728                 WHEN OTHER MOVE '?' ' TO WW-TYPE-LABEL
729             END-EVALUATE
730             DISPLAY '   '
731                 DB-FIRST-NAME(WW-FOUND-IDX) ' '
732                 DB-LAST-NAME(WW-FOUND-IDX)
733                 '   ID#' DB-CLIENT-ID(WW-FOUND-IDX)
734                 '   ' WW-STATUS-LABEL
735                 '   ' WW-TYPE-LABEL
736             .
737
738         *-----*
739         9200-LOG-TRANSACTION.
740             ADD 1 TO TX-TOTAL-COUNT
741             COMPUTE WW-TX-SLOT =
742                 FUNCTION MOD(TX-TOTAL-COUNT - 1, BR-MAX-TX-LOG) + 1
743             MOVE TX-NEXT-ID          TO TX-ID(WW-TX-SLOT)
744             ADD 1 TO TX-NEXT-ID
745             MOVE WW-DATE-SIM          TO TX-DATE(WW-TX-SLOT)
746             MOVE WW-TIME-SIM          TO TX-TIME(WW-TX-SLOT)
747             MOVE DB-CLIENT-ID(WW-FOUND-IDX)
748                 TO TX-CLIENT-ID(WW-TX-SLOT)
749             MOVE WW-CURRENT-OP        TO TX-TYPE(WW-TX-SLOT)
750             MOVE WW-AMOUNT            TO TX-AMOUNT(WW-TX-SLOT)
751             MOVE DB-BALANCE(WW-FOUND-IDX)
752                 TO TX-BAL-AFTER(WW-TX-SLOT)
753             MOVE WW-RESULT            TO TX-RESULT(WW-TX-SLOT)
754             IF WW-RESULT NOT = 'OK'
755                 MOVE WW-ERROR-MSG TO TX-REJECT-REASON(WW-TX-SLOT)
756             ELSE
757                 MOVE SPACES          TO TX-REJECT-REASON(WW-TX-SLOT)
758             END-IF
759             .

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760
761      *-----*
762      9300-DISPLAY-FULL-RECORD.
763      DISPLAY WD-DASH-LINE
764      DISPLAY ' == FULL ACCOUNT RECORD =='
765      DISPLAY WD-DASH-LINE
766      DISPLAY ' CLIENT ID      : '
767      DB-CLIENT-ID(WW-FOUND-IDX)
768      DISPLAY ' LAST NAME      : '
769      DB-LAST-NAME(WW-FOUND-IDX)
770      DISPLAY ' FIRST NAME     : '
771      DB-FIRST-NAME(WW-FOUND-IDX)
772      MOVE DB-BALANCE(WW-FOUND-IDX) TO WD-BALANCE
773      DISPLAY ' BALANCE        : ' WD-BALANCE
774      DISPLAY ' STATUS         : '
775      DB-STATUS(WW-FOUND-IDX)
776      ' (A=ACTIVE F=FROZEN C=CLOSED)'
777      DISPLAY ' ACCOUNT TYPE   : '
778      DB-ACCT-TYPE(WW-FOUND-IDX)
779      ' (R=REGULAR V=VIP)'
780      DISPLAY ' DATE OPENED    : '
781      DB-OPEN-DATE(WW-FOUND-IDX)
782      DISPLAY ' TOTAL TX COUNT : '
783      DB-TX-COUNT(WW-FOUND-IDX)
784      MOVE DB-DAILY-WD(WW-FOUND-IDX) TO WD-DAILY-WD
785      DISPLAY ' WITHDRAWN TODAY: ' WD-DAILY-WD
786      MOVE DB-DAILY-DEP(WW-FOUND-IDX) TO WD-DAILY-DEP
787      DISPLAY ' DEPOSITED TODAY: ' WD-DAILY-DEP
788      DISPLAY ' LAST TX DATE   : '
789      DB-LAST-TX-DATE(WW-FOUND-IDX)
790      DISPLAY ' LAST TX TYPE   : '
791      DB-LAST-TX-TYPE(WW-FOUND-IDX)
792      ' (D=DEPOSIT W=WITHDRAWAL)'
793      MOVE DB-LAST-TX-AMT(WW-FOUND-IDX) TO WD-AMOUNT
794      DISPLAY ' LAST TX AMOUNT : ' WD-AMOUNT
795      DISPLAY WD-DASH-LINE
796      DISPLAY ' -- APPLICABLE LIMITS -- '
797      IF DB-VIP(WW-FOUND-IDX)
798      MOVE BR-VIP-MAX-DEP-SINGLE TO WD-AMOUNT
799      DISPLAY ' MAX SINGLE DEP : ' WD-AMOUNT ' (VIP)'
800      MOVE BR-VIP-MAX-WD-SINGLE TO WD-AMOUNT
801      DISPLAY ' MAX SINGLE WD  : ' WD-AMOUNT ' (VIP)'
802      MOVE BR-VIP-MAX-DEP-DAILY TO WD-AMOUNT
803      DISPLAY ' DAILY DEP LIMIT: ' WD-AMOUNT ' (VIP)'
804      MOVE BR-VIP-MAX-WD-DAILY TO WD-AMOUNT
805      DISPLAY ' DAILY WD LIMIT : ' WD-AMOUNT ' (VIP)'
806      ELSE
807      MOVE BR-MAX-DEP-SINGLE TO WD-AMOUNT
808      DISPLAY ' MAX SINGLE DEP : ' WD-AMOUNT ' (REG)'
809      MOVE BR-MAX-WD-SINGLE TO WD-AMOUNT
810      DISPLAY ' MAX SINGLE WD  : ' WD-AMOUNT ' (REG)'
811      MOVE BR-MAX-DEP-DAILY TO WD-AMOUNT
812      DISPLAY ' DAILY DEP LIMIT: ' WD-AMOUNT ' (REG)'
813      MOVE BR-MAX-WD-DAILY TO WD-AMOUNT
814      DISPLAY ' DAILY WD LIMIT : ' WD-AMOUNT ' (REG)'
815      END-IF
816      MOVE BR-MIN-BALANCE TO WD-AMOUNT
817      DISPLAY ' MIN BALANCE REQ: ' WD-AMOUNT
818      DISPLAY WD-DASH-LINE
819      .
820
821      *=====*
822      * END-OF-DAY RESET *
823      *=====*
824
825      9500-RESET-DAILY-LIMITS.
826      DISPLAY WD-DASH-LINE
827      DISPLAY ' RESET DAILY LIMITS - NEW BUSINESS DAY'
828      DISPLAY WD-DASH-LINE

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829      DISPLAY '  CONFIRM RESET? (Y/N): ' WITH NO ADVANCING
830      ACCEPT WW-MENU-CHOICE
831      IF WW-MENU-CHOICE NOT = 'Y' AND
832         WW-MENU-CHOICE NOT = 'y'
833         DISPLAY '  RESET CANCELLED.'
834         EXIT PARAGRAPH
835      END-IF
836      PERFORM VARYING WW-LOOP-CTR FROM 1 BY 1
837         UNTIL WW-LOOP-CTR > DB-CLIENT-COUNT
838         MOVE ZEROS TO DB-DAILY-WD(WW-LOOP-CTR)
839         MOVE ZEROS TO DB-DAILY-DEP(WW-LOOP-CTR)
840      END-PERFORM
841      DISPLAY '  DAILY LIMITS RESET FOR ALL '
842         DB-CLIENT-COUNT ' ACCOUNTS.'
843      DISPLAY '  NEW BUSINESS DAY COMMENCED.'
844      .
845
846      *=====*
847      * TRANSACTION LOG DISPLAY *
848      *=====*
849
850      9600-SHOW-TX-LOG.
851      DISPLAY WD-DASH-LINE
852      DISPLAY ' TRANSACTION LOG - LAST 20 ENTRIES'
853      DISPLAY WD-DASH-LINE
854      IF TX-TOTAL-COUNT = 0
855         DISPLAY ' NO TRANSACTIONS RECORDED YET.'
856         EXIT PARAGRAPH
857      END-IF
858      DISPLAY ' TXID      DATE      CLIENT OP  AMOUNT'
859         '          RESULT'
860      DISPLAY WD-DASH-LINE
861      COMPUTE WW-PAGE-START = TX-TOTAL-COUNT - 19
862      IF WW-PAGE-START < 1
863         MOVE 1 TO WW-PAGE-START
864      END-IF
865      PERFORM VARYING WW-LOOP-CTR
866         FROM WW-PAGE-START BY 1
867         UNTIL WW-LOOP-CTR > TX-TOTAL-COUNT
868         COMPUTE WW-TX-SLOT =
869             FUNCTION MOD(WW-LOOP-CTR - 1, BR-MAX-TX-LOG) + 1
870         MOVE TX-AMOUNT(WW-TX-SLOT) TO WD-AMOUNT
871         DISPLAY ' '
872             TX-ID(WW-TX-SLOT) ' '
873             TX-DATE(WW-TX-SLOT) ' '
874             TX-CLIENT-ID(WW-TX-SLOT) ' '
875             TX-TYPE(WW-TX-SLOT) ' '
876             WD-AMOUNT ' '
877             TX-RESULT(WW-TX-SLOT)
878      END-PERFORM
879      DISPLAY WD-DASH-LINE
880      .
881
882      *=====*
883      * END PROGRAM *
884      *=====*
885
886      9999-END-PROGRAM.
887      DISPLAY WD-SEP-LINE
888      DISPLAY ' SESSION CLOSED'
889      DISPLAY ' TRANSACTIONS THIS SESSION : ' TX-TOTAL-COUNT
890      DISPLAY ' BANK OF COBOL - HAVE A GOOD DAY'
891      DISPLAY WD-SEP-LINE
892      .
893
894      *=====*
895      * END OF PROGRAM: BANKING-SYSTEM *
896      *=====*

```